Efficiency and energy audits

National 5 · Engineering contexts & challenges · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Calculate the efficiency of the bulb as a light source.

- E_{in} = 12 × 5 × 600 = 36 000 J
- E_h = 4180 × 0.5 × 12 = 25 080 J (wasted)
- Light = 36 000 − 25 080 = 10 920 J
- Efficiency = 10 920 ÷ 36 000 = 30%

2 Water in a hydroelectric station falls 500 m at a rate of 5000 kg per second. The energy transfer is 65% efficient. Calculate the useful energy produced each second.

- E_{in} = mgh = 5000 × 9.8 × 500 = 24 500 000 J
- E_{out} = 0.65 × 24 500 000 = 15 925 000 J = 16 MJ

3a) Calculate the heat energy absorbed by the water.

- ΔT = 230 °C
- E_h = 4180 × 100 × 230 = 96 140 000 J = 96 MJ

3b) For every 15 MJ of heat from the ground the plant produces 5.34 MJ of electricity. Calculate its efficiency.

- η = 5.34 ÷ 15 = 0.356 = 36%

4 A deep fat fryer uses 2 kJ of electrical energy each second. It heats 1.5 kg of oil (c = 2400 J/kg K) from 20 °C to 80 °C in 6 minutes. Calculate the efficiency of the fryer.

- E_h = 2400 × 1.5 × 60 = 216 000 J
- E_{in} = 2000 × 360 = 720 000 J
- η = 216 000 ÷ 720 000 = 0.30 = 30%

5 A heater on a 240 V supply draws 8 A and is 75% efficient. How long will it take to heat 1.5 litres of water from 20 °C to 90 °C?

- E_h = 4180 × 1.5 × 70 = 438 900 J (useful output)
- E_{in} = 438 900 ÷ 0.75 = 585 200 J
- t = E_{in} ÷ (VI) = 585 200 ÷ (240 × 8) = 305 s (about 5 minutes)

6a) Complete the energy lost column.

- Motor 1.5 kJ; gearbox 2.7 kJ; winding mechanism 1.7 kJ

6b) Calculate the total energy lost.

- 1.5 + 2.7 + 1.7 = 5.9 kJ

6c) Complete the sentence: "Because the ..... is always greater than the ..... the efficiency will always be less than 100%."

- input energy ... output energy

6d) Calculate the efficiency of the ski tow.

- η = 9.1 ÷ 15 = 0.61 = 61%

7a) Calculate the electrical energy supplied.

- E_e = 230 × 22 × 16 = 80 960 J = 81 kJ

7b) Calculate the heat energy transferred to the water.

- E_h = 4180 × 0.14 × 71 = 41 549 J = 42 kJ

7c) Calculate the efficiency of the machine.

- η = 41 549 ÷ 80 960 = 0.51 = 51%